A tangent-groupoid formulation of the quantum construction of Newton motion

Result, 2026-10-27. The tangent groupoid carries a continuous field of observable algebras whose positive-parameter fibers are quantum and whose zero fiber is classical. After fixing an action unit S_* and setting $\hbar = \epsilon S_*$, the finite quantum products and classical initial-value construction of the quantum product theorem define compatible states on these fibers. On every compact time interval, their values on smooth Weyl observables converge to the classical point state with an explicit $O(\hbar + |\pi|)$ bound, for **every joint schedule** with $\hbar, |\pi| \to 0$.

The result is a state-level limit for the specified coherent preparations and smooth potentials with bounded derivatives of order at least two, including the third derivative used for the rate. The quartic extension in the quantum product theorem uses cutoff Hamiltonians; it does not establish this estimate for uncut quartic quantum evolution. The result neither makes the finite-product propagator uniformly close to the exact quantum propagator as $\hbar \to 0$, nor makes a quantum superposition a single classical trajectory. The groupoid parameter is a geometric deformation parameter until the identification $h = \epsilon S_*$ is chosen. The time mesh is an additional scale.

The wall construction treats boundary domains and impact separately; the whole-line theorem here does not cover that problem. The classical construction and quantum product theorem supply the dynamics used below.

The groupoid and strict quantization are established: in particular, Cariñena, Clemente-Gallardo, Follana, Gracia-Bondía, Rivero and Várilly, Connes' Tangent Groupoid and Strict Quantization (abstract and the prior repository companion), already includes the user's work. Landsman–Ramazan's continuous-field construction (abstract) supplies the general Lie-groupoid setting. The following state estimate applies the quantum product theorem's finite-slice bounds to that established structure; no literature priority claim is made.

1. What the parameter joins

For $M = \mathbb{R}$, the tangent groupoid is

$$\mathbb{T}M = (M \times M \times (0,1]) \;\sqcup\; (TM \times \{0\}). \tag{1}$$

At $\epsilon > 0$, pair arrows compose as $(x,y,\epsilon)(y,z,\epsilon) = (x,z,\epsilon)$; at zero, $(q,v,0)(q,w,0) = (q, v+w, 0)$. A local chart near the zero fiber is

$$(q,v,\epsilon) \longmapsto (q + \epsilon v/2,\; q - \epsilon v/2,\; \epsilon). \tag{2}$$

Thus $v = (x-y)/\epsilon$ is a **tangent coordinate**, not automatically a physical velocity. A physical time interpretation requires a separate identification of units and scaling.

The groupoid C^*-algebra is a continuous field with fibers

$$\mathcal{A}_\epsilon \simeq \mathcal{K}(L^2(\mathbb{R})) \quad (\epsilon > 0), \qquad \mathcal{A}_0 \simeq C_0(T^*\mathbb{R}). \tag{3}$$

At zero, Fourier transformation takes tangent-vector convolution to pointwise multiplication of functions of (q,p). The positive fibers are abstractly isomorphic, but the continuous field specifies how families of their elements approach a classical observable.

Choose an action unit $S_* > 0$ and set

$$\hbar = \epsilon S_*. \tag{4}$$

For $a \in C_c^\infty(T^*\mathbb{R})$, Weyl quantization has kernel

$$K_\epsilon^a(x,y) = \frac{1}{2\pi\epsilon S_*} \int_{\mathbb{R}} e^{i(x-y)p/(\epsilon S_*)} a\!\left(\frac{x+y}{2}, p\right) dp. \tag{5}$$

In chart (2),

$$\epsilon K_\epsilon^a(q+\epsilon v/2, q-\epsilon v/2) = \frac{1}{2\pi S_*}\int e^{ivp/S_*} a(q,p)dp. \tag{6}$$

The rescaled kernel is the inverse Fourier transform of the zero-fiber symbol. The established strict-quantization construction makes $\epsilon \mapsto \mathrm{Op}^W_{\epsilon S_*}(a)$, together with a at zero, a continuous section. Equation (6) displays the chart and normalization in this flat case; it is not a new proof of the general continuous-field theorem.

2. The dynamics and the unbounded objects

For the potential and hypotheses of the quantum product theorem define

$$H_\hbar = P^2/(2m) + V(Q), \qquad U_\hbar(t) = e^{-itH_\hbar/\hbar}. \tag{7}$$

The $H_\hbar$ are unbounded operators and the $U_\hbar(t)$ are generally multipliers, not elements of the compact-operator fibers. At $\hbar > 0$ their adjoint action defines automorphisms of the compact operators:

$$\alpha_\hbar^t(A) = U_\hbar(t)^* A U_\hbar(t). \tag{8}$$

The quantum product theorem constructs the unique classical phase-space flow $\Phi_t(q,p)$ before it is used here. On the zero fiber put

$$\alpha_0^t(a) = a \circ \Phi_t. \tag{9}$$

This is classical Hamiltonian evolution with $H_{\mathrm{cl}} = p^2/(2m) + V(q)$.

For $a \in C_c^\infty(T^*\mathbb{R})$, Weyl–Egorov compatibility for the smooth subquadratic potentials of the quantum product theorem gives, for fixed T,

$$\sup_{|t|\le T} \|\alpha_\hbar^t(\mathrm{Op}_\hbar^W(a)) - \mathrm{Op}_\hbar^W(a\circ\Phi_t)\| \le C_{a,T}\hbar^2. \tag{10}$$

Here is the relevant estimate. The finite-time flow has bounded derivatives on compact sets under the stated hypotheses, so $a_s = a \circ \Phi_s$ is smooth, compactly supported, and has uniform symbol bounds for $|s| \le T$. The kinetic commutator is exact. The potential-commutator remainder after removing its classical first-order term has kernel

$$\frac{i}{\hbar}\left[V(x) - V(y) - (x-y)V'\!\left(\frac{x+y}{2}\right)\right] K_\hbar^{a_s}(x,y). \tag{10a}$$

The bracket is bounded by $\|V'''\|_\infty |x-y|^3/24$. Rapid decay of the Fourier transform of a_s in its momentum variable, the change $x - y = hz$, and Schur's test bound the operator norm by $C_{a,T}\hbar^2$. Integrating this commutator remainder by Duhamel's formula gives (10). This uses the standard Weyl calculus and the flow already constructed in the quantum product theorem; it is not being used to prove that flow exists.

3. A quantitative state on every fiber

Use the coherent packet $\psi_\hbar^{q_0,p_0}$ of equation (16) in the quantum product theorem. For a partition π, let $\psi_{\hbar,\pi,j}$ be the state after its first j kick–drift cells, and let $z_{\hbar,\pi}(t_j) = (q_j, p_j)$ be its means. Denote the constructed classical state by $z(t) = \Phi_t(q_0,p_0)$. Define

$$I_a = (2\pi)^{-2}\int_{\mathbb{R}^2} |\widehat{a}(\xi,\eta)|\,(|\xi| + m\omega|\eta|)\,d\xi\,d\eta < \infty. \tag{11}$$

Here $\widehat{a}$ is the ordinary Fourier transform in (q,p). Also put

$$J_a = \frac{1}{2(2\pi)^2}\int_{\mathbb{R}^2} |\widehat{a}(\xi,\eta)|\,(|\xi| + m\omega|\eta|)^2\,d\xi\,d\eta < \infty. \tag{11a}$$

Using the exact Fourier representation of Weyl quantization,

$$\mathrm{Op}_\hbar^W(a) = (2\pi)^{-2} \int \widehat{a}(\xi,\eta) e^{i(\xi Q+\eta P)} d\xi\, d\eta, \tag{12}$$

set $A = \xi(Q-q_j)+\eta(P-p_j)$. The scalar phase $e^{i(\xi q_j+\eta p_j)}$ factors out, $\langle A\rangle_j = 0$, and spectral calculus gives $|\langle e^{iA}\rangle_j - 1| \le \frac{1}{2}\langle A^2\rangle_j$. Consequently,

$$|\langle \mathrm{Op}_\hbar^W(a)\rangle_j - a(q_j,p_j)| \le J_a \left[\mathrm{sd}(Q)_j + \frac{\mathrm{sd}(P)_j}{m\omega}\right]^2. \tag{13}$$

Indeed $\langle A^2\rangle_j = \|A\psi_{\hbar,\pi,j}\|^2$ and the triangle inequality bounds $\|A\psi_{\hbar,\pi,j}\|$ by the weighted sum of the two standard deviations. No Gaussian-shape assumption is needed after the first step. Fourier inversion also gives $|a(z)-a(z')| \le I_a \|z-z'\|_\omega$.

The quantum product theorem's estimates (19) and (26) therefore give, at every partition vertex $t_j \in [0,T]$,

$$\left|\langle \psi_{\hbar,\pi,j}, \mathrm{Op}_\hbar^W(a)\psi_{\hbar,\pi,j}\rangle - a(z(t_j))\right| \le J_a C^2 e^{2\beta T}\hbar + I_a G_T(C_\delta|\pi| + C_h h). \tag{14}$$

The constants $C, \beta, G_T, C_\delta, C_\hbar$ are explicitly defined in that document and depend on the common $\hbar_0, T, V, m, \omega$ and initial data, not on $\hbar$ or the number of cells. No differentiability of an unbounded quantum expectation is used in this step.

With $\hbar = \epsilon S_*$, choose any $\alpha > 0$ and a time unit $T_* > 0$ so that $|\pi_\epsilon| \le T_* \epsilon^\alpha$. Equation (14) yields

$$\langle \mathrm{Op}_{\epsilon S_*}^W(a)\rangle_{\pi_\epsilon} \longrightarrow a(\Phi_t(q_0,p_0)) \quad \text{at vertices tending to } t, \tag{15}$$

with error $O(\epsilon+\epsilon^\alpha)$ at vertices equal to t. Such a vertex can be inserted in each partition. There is no constraint here on $|\pi_\epsilon|/\hbar$.

This means that the vector states on the positive fibers approach the point-evaluation state on the zero fiber, for the specified family of preparations. For an arbitrary continuous section A, approximate $A(0) \in C_0(T^*\mathbb{R})$ uniformly by compactly supported smooth symbols. Continuity of the field and the norm-one bound on states extend (15) from these symbols to A. At fixed $\hbar$, taking the mesh limit recovers the exact unitary state as in the quantum product theorem. Both routes thus reach the same zero-fiber state; equality or closeness of the two quantum vectors along an arbitrary joint schedule has **not** been established.

4. A precise two-scale test on Euclidean increments

The preceding theorem concerns real-time operator states. It does not claim that Euclidean sample paths are real-time trajectories. The following free Gaussian calculation shows why the groupoid coordinate, the action parameter and the duration must not be conflated.

Consider a drifted free reference increment

$$Y - X = u\delta + \sqrt{\frac{\hbar\delta}{m}}\, Z, \qquad Z \sim N(0,1). \tag{16}$$

Set $\hbar = \epsilon S_*$ and $\delta = T_*\epsilon^\alpha$ with $\alpha > 0$. The physical-time difference quotient is

$$\frac{Y-X}{\delta} = u + \sqrt{\frac{S_*}{mT_*}}\, \epsilon^{(1-\alpha)/2} Z, \tag{17}$$

whereas the tangent-groupoid chart variable is

$$v_\epsilon = \frac{Y-X}{\epsilon} = uT_*\epsilon^{\alpha-1} + \sqrt{\frac{S_*T_*}{m}}\, \epsilon^{(\alpha-1)/2} Z. \tag{18}$$

- If $0 < \alpha < 1$, the physical-time quotient concentrates at u, but v_ϵ is not tight.
- If $\alpha = 1$, v_ϵ converges in law to a nondegenerate Gaussian of mean uT_* and variance S_*T_*/m.
- If $\alpha > 1$, $v_\epsilon \to 0$ in probability, while the physical-time quotient is not tight.

This is an exact diagnostic for the Euclidean Gaussian reference, not a probability law for real-time Feynman paths. It also does not contradict (15): a groupoid observable is obtained after the Fourier and operator construction, whereas (18) examines a single sampled Euclidean increment.

5. Where blocking sits in the groupoid picture

For each $\epsilon > 0$ the quantum product theorem supplies a complete tower $\rho_{\epsilon,n}$ of normalized conditional densities on dyadic time samples. The relation

$$\rho_{\epsilon,n} = \mathbb{E}_{0,\epsilon}[\rho_{\epsilon,n+1} \mid \mathcal{F}_n] \tag{19}$$

is exact and retains all generated interactions. The integer n counts an integration/coarse-graining operation; ϵ labels the deformation field (4). The conditional maps are generally not invertible, so they are not automatically arrows of the tangent groupoid.

Equation (15) connects these scales for the real-time **observable states** even when they are tied by a single chosen schedule. It does not follow from the $1/Z_\epsilon$ Euclidean bound, which need not be uniform in ϵ, and it does not promote Euclidean minimization into real-time Newton dynamics.

The zoom action on the tangent groupoid is a geometric rescaling of near-diagonal arrows. It is useful for organizing a comparison with RG, but it is not by itself an integration of fluctuations or a theorem that the physical action constant runs. Likewise, the real-time Hamilton principal action composes by stationary elimination on a specified noncaustic branch; infimal convolution is appropriate for a genuine minimizing Euclidean problem or a separately proved special case. The kernel calculation below makes this distinction explicit.

6. Regulators and a normalized blocking test

The connection is already part of Rivero's work. The appendix to his 1997 tangent-groupoid note (pp. 7–8, passages read) constructs a **toy RG triangle**: a dilation rescales a bare sequence approaching the tangent boundary to a chosen nonzero reference slice, with limits at different slices related by rescalings. His 1998 Feynman-formula proposal (pp. 2–3, equations (5)–(10), passages read) similarly distinguishes two regulators and calls its controlled objects bare, renormalized and dressed series. Its control transformation is left unspecified; its control constant is arbitrary. The 1999 strict-quantization paper with Rivero and coauthors supplies the groupoid quantization framework. These are the provenance of the present question, not a claim that the recent calculation invented the groupoid/RG connection.

The different appearances of epsilon need an explicit dictionary. Let ϵ_G be a dimensionless groupoid deformation parameter; in the specified state formulation, choosing $\epsilon_G = \hbar/S_0$ assigns the dimensionful phase constant to that parameter. Let $\eta = |\pi|/T_0$ be a dimensionless mesh scale, and let α_π be an action-dimensional bare phase regulator. A possible control map, already recorded in the 1998 test, is

$$S_\pi^{\mathrm{bare}} = Z_S(\eta) S_\pi, \qquad \hbar_{\mathrm{eff}} = \frac{\alpha_\pi}{Z_S(\eta)}.$$

For example, the *specified* choices $Z_S(\eta) = \eta$ and $\alpha_\pi = \eta\hbar$ give $e^{iS_\pi^{\mathrm{bare}}/\alpha_\pi} = e^{iS_\pi/\hbar}$ at every mesh. This algebra illustrates how a vanishing bare regulator can coexist with a fixed positive effective action; substituting the same regulator into an unscaled action would give a different phase.

An exact Gaussian blocking test. Let $m > 0$, $u \in \mathbb{R}$ and let $a \geq 0$ be an action-dimensional parameter. Define a normalized transition kernel by

$$K^u_{a,t}(x, dy) = \mathcal{N}(x + ut, at/m)(dy), \qquad t > 0,$$

with zero variance interpreted as a Dirac measure. For $a = \hbar > 0$ and $u = 0$ this is the free Euclidean quantum kernel; arbitrary u gives the drifted Gaussian reference, not a real-time path measure. Integrating out an intermediate position preserves a exactly:

$$K^u_{a,t} K^u_{a,v} = K^u_{a,t+v}.$$

If time and position are then rescaled by $t_{\text{old}} = r t_{\text{new}}$, $x_{\text{old}} = s x_{\text{new}}$ with $r, s > 0$, the normalized kernel obeys

$$K^u_{a,rt}(sx, sB) = K^{ru/s}_{ra/s^2,t}(x, B) \quad \text{for every Borel set } B.$$

These identities follow from adding independent Gaussian increments and rescaling their means and variances; they include $a = 0$. Under the diffusive choice $r = b$, $s = \sqrt{b}$, the action coefficient stays fixed. When $u = 0$, every $a \geq 0$ is a fixed point of this particular normalized block-and-rescale map. At $a = 0$, retaining u gives the deterministic free Newton motion $x(t) = x(0) + ut$. Thus this exact free RG control neither selects a positive action nor excludes the classical branch. It is an illustrative Gaussian comparison, not a result about all interacting blocking maps or complete record instruments; compare the free increment and the action-normalization identity.

The two-regulator bridge and a residual action. For the free Gaussian endpoint bridge on a partition with $d = N - 1$ interior points, whitening the quadratic action gives $2\Delta S/a \sim \chi^2_d$ and hence

$$\mathbb{E}\Delta S = \frac{da}{2}, \qquad \operatorname{Var}(\Delta S) = \frac{da^2}{2}.$$

If the maximum cell length and a_N tend to zero while $da_N \to \ell$, the bridges can be coupled to converge uniformly almost surely to the classical chord, while $\Delta S \to \ell/2$ in L^2. This uses one Brownian bridge and its polygonal interpolations. For equal cells $\tau = T/N$ and $a_N = b\tau$, the supplied b has energy units and $\ell = bT$. The positive residual is selected by the scaling; at every finite mesh the action distribution has values arbitrarily near zero.

The separately controlled reference family fixes $m_R, \kappa_R > 0$ and chooses $m_B = \alpha_\pi m_R/\kappa_R$, so its normalized free kernels are preserved as $\alpha_\pi \to 0$; the bare mass then tends to zero. This is different from the preceding fixed-mass limit and does not derive the reference action. The regulator manuscript, Proposition 2 and its running-coefficient section, contains the full arbitrary-partition proof and source reading; the PDF, B08 and its review retain the detailed record. The manuscript also gives a bounded-speed shrinking sine example with a fixed action defect and growing acceleration, and a separate scalar stationary-phase/delta-weight calculation. Those examples motivate testing which independent force or derivative bounds suppress such a defect; they do not make the defect universal or identify it with $\hbar$.

The open RG problem is to construct a compatible control or blocking map for the actual normalized kernels, states and generated interactions, including the increments and all required limits, and then determine whether its effective action has a universal physical identification. Geometric zoom, coarse-graining and running parameters can participate in one construction, but a geometric dilation alone does not establish that construction or force $\hbar$ to run. The corrected groupoid comparison and scale correspondence keep these claims and their corrections distinct.

Geometric zoom and kernel composition

The adiabatic zoom acts by $(x, y, \epsilon) \mapsto (x, y, \lambda^{-1}\epsilon)$ and $(x, v, 0) \mapsto (x, \lambda v, 0)$. Debord–Skandalis 2014 and van Erp–Yuncken 2019 (metadata read) relate this geometry to pseudodifferential kernels homogeneous modulo smoothing terms; the zero-fiber restriction gives their principal

symbols. This source result identifies geometric structure, not a Wilsonian integration or a law for physical $\hbar$. Earlier identifications of all three were too broad.

The user's 2003 note (Rivero, "Flashes of noncommutativity", arXiv:math/0302285, full read) introduces the groupoid over configuration space with arrows (x, y, u), u an elapsed time, and law $(x, y, u) \circ (y, z, v) = (x, z, u + v)$. Its convolution algebra has product $(AB)(x, z, t) = \iint A(x, y, r)\, B(y, z, t - r)\, dy\, dr$; Fourier transformation in t gives, with $\varepsilon \sim 1/\hat{t}$, the product of the $\varepsilon > 0$ part of the tangent groupoid, so the latter's algebra sits inside the former's (the 2003 note's second flash). Refining a trajectory is factoring an arrow (x, z, u) through intermediate points and times, which is Newton's insertion of $t_{2.6}$.

For endpoint functions define min-plus convolution by

$$(A \otimes B)(x, z) := \inf_y \{A(x, y) + B(y, z)\}.$$

It eliminates the intermediate endpoint; it does not add the action on each individual composable pair of arrows.

Proposition 1. For $M > 0$, constant force F, and $u > 0$, set

$$S_u(x, y) = \frac{M(y - x)^2}{2u} + \frac{Fu}{2}(x + y) - \frac{F^2u^3}{24M}.$$

Then $(S_u)_{u>0}$ is a semigroup of kernels under min-plus convolution: $S_r \otimes S_v = S_{r+v}$ for $r, v > 0$. This is the corrected action ℓ_u of the refinement note, equation (2). The corresponding constant-force unitary propagators satisfy $U_{r+v} = U_r U_v$; their kernels compose by integration over the intermediate position, in the oscillatory/distributional sense where the real-time integrals are not absolutely convergent.

Proof. The sum $S_r(x, y) + S_v(y, z)$ is a strictly convex quadratic in y with unique minimizer

$$y_* = \frac{vx + rz}{r + v} - \frac{Frv}{2M}.$$

Substitution gives $S_{r+v}(x, z)$. Equivalently, y_* is the value at time r of the action-minimizing constant-force path joining x to z. The unitary law follows from the self-adjoint Hamiltonian; its kernel statement is operator composition. □

The min-plus law is not an ordinary groupoid morphism. For free motion with $M = 1$, $S_1(0, 1) + S_1(1, 0) = 1$, whereas $S_2(0, 0) = 0$. The equality holds only after minimizing over the intermediate point ($y = 0$ in this example).

The displayed S is the exact discrete Lagrangian of discrete mechanics (Marsden and West 2001, metadata), and discrete Lagrangian mechanics on groupoids is Weinstein's (1996) (metadata). The kick–drift–kick cell $S_\hbar = K_{Fh/2} D_h K_{Fh/2}$ of the refinement note is the variational (Störmer–Verlet) integrator of the trapezoidal discrete Lagrangian $L_\hbar$, and the cubic counterterm turns $L_\hbar$ into the exact one. The laws of Proposition 1 can be related through a Euclidean Laplace limit: under coercivity and regularity conditions, $-\hbar \log \int e^{-S/\hbar} dy \to \inf_y S$ as $\hbar \to 0$ (Litvinov 2005, metadata). This does not turn an oscillatory real-time integral into a probability integral, nor make every stationary real-time branch a global minimum.

7. Consequence for STATE

The constructed $0 + 1$ quantum family, the semiclassical Newton flow and the user's tangent-groupoid framework now meet in a concrete state theorem with a joint mesh/action error bound. The strong operator product limit remains a fixed-$\hbar$ theorem; the Euclidean density tower and the real-time coherent states have distinct roles. No positive universal action scale or four-dimensional Yang–Mills result follows.

Targeted advisor check (GPT-6.1 Sol, 2026-10-27). An internal mathematical review checked the chart and Fourier normalizations, the state and Gaussian-scale bounds, the extension

to continuous sections, and the estimate (10). It is not independent peer review of the underlying continuous-field theorem or of the literature. The centered $O(\hbar)$ state improvement in §3 is a subsequent written derivation and was not part of that earlier check.